MACHINE LEARNING AND AI FOR ANALYTICAL CHEMISTRY

Part 1: Foundations of Machine Learning and AI

M2 Sciences analytiques - 6 h lectures + 1 h tutorial

Lennart Martens - Jerome Pansanel

WHAT THIS PART COVERS

Roadmap for Part 1

1

Concepts

What ML and AI mean, the learning settings, features, and the modelling workflow

2

Classical models

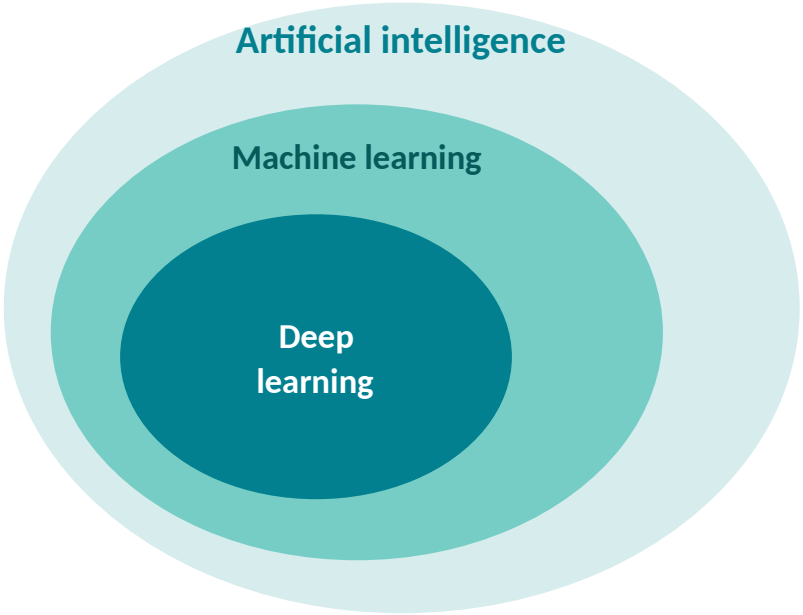
Decision trees and how they are induced, random forests, SVM/SVR, k-means

3

Deep learning

Neural networks, CNNs, transformers and attention, with GNNs and autoencoders in brief

AI, machine learning, deep learning



Three nested ideas

AI is the broad goal of machine intelligence.

ML fits a function to data instead of hand-coding rules.

Deep learning is ML with many-layered neural networks.

Example: a Beer-Lambert calibration is a physical rule; predicting LC retention time is ML; DeepLC is its deep-learning version.

CONCEPTS

Why this matters for analytical chemistry

Rich measurements

Chromatograms, mass, IR, Raman and NMR spectra, ion mobility: large, structured, high-dimensional data.

Hard-to-write rules

Models can capture relationships too complex to express as explicit physical or expert rules.

New capabilities

Predict retention, fragmentation and CCS; recognise signals; flag anomalies (Part 4).

A real risk

A model can look like it works while having learned the wrong thing (Part 3).

Three learning settings

Supervised

Examples carry known answers (labels). Learn input to answer.

Example: *GC retention index from descriptors (regression); Raman polymorph A vs B (classification).*

Unsupervised

No labels. Find structure on its own.

Example: *Cluster olive-oil GC-MS profiles to see if they separate by geographic origin.*

Reinforcement

Learn by trial and error against a reward.

Example: *Tune MS collision energy to maximise confident identifications. Not a course focus.*

CONCEPTS

Features: the central idea

A feature is a measurable property of a sample, as a number. A dataset is a table: rows are samples, columns are features, plus the target. A spectrum becomes a vector with one absorbance per wavenumber (an IR spectrum is about 1800 numbers).

sample	alcohol	flavanoids	proline	-> target
wine 1	14.2	3.06	1065	cultivar 0
wine 2	13.2	2.76	1050	cultivar 0
wine 3	12.4	0.92	562	cultivar 1

Key point: the quality and origin of the features usually matter more than the choice of algorithm (Part 2).

The machine learning workflow

1. **Define the task**
2. **Understand the data**
3. **Build features**
4. **Split the data**
5. **Train**
6. **Evaluate**
7. **Interpret & iterate**
8. **Deploy & monitor**

Running example: flag out-of-spec batches from NIR spectra. Steps 4 and 6 (green) decide whether you can trust the result (Part 3).

CONCEPTS

Two cautions to start with

When NOT to use ML

- A validated physical model already exists
- Very few samples
- Full transparency is required
- A single rule solves the problem
- Always beat a trivial baseline first

Example: *a linear Beer-Lambert fit beats a neural net for single-analyte quantitation.*

Overfitting (preview of Part 3)

A too-flexible model memorises noise and fails on new data. Underfitting is the opposite: too simple to capture the pattern.

Good scores on data the model has already seen prove almost nothing.

Example: *a classifier that secretly reads the measurement batch, not the chemistry.*

Classical machine learning

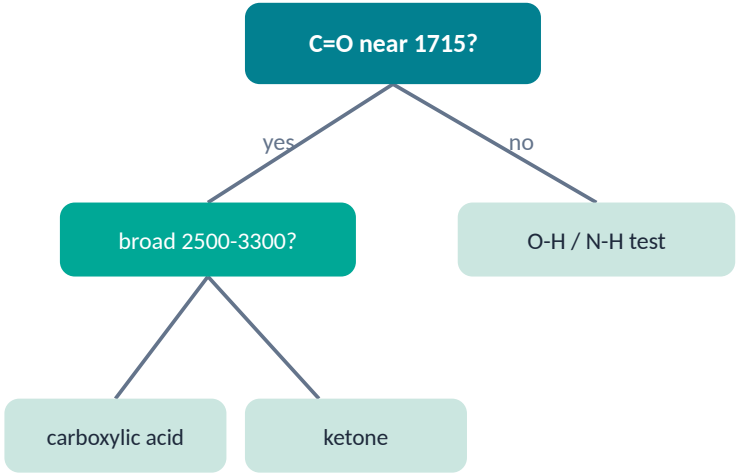
Decision trees and their induction - random forests - SVM/SVR - k-means

Decision trees

A sequence of yes/no questions on the features.

+ readable, no scaling, handles non-linearities and interactions.

- a deep single tree overfits badly (controlled by pruning, next slides).



Example: identify a functional group from an IR spectrum by splitting on diagnostic absorption bands.

How a decision tree is induced

The induction algorithm

1. Start with all samples at the root.
2. Try every feature and every split point.
3. Score each split by impurity reduction.
4. Keep the best split; partition the samples.
5. Recurse on each child node.
6. Stop when pure or a limit is reached.

Greedy: *the locally best split now, with no look-ahead.*

Scoring a split

Entropy H = - Sum p log2 p

Gini G = 1 - Sum p^2

Information gain

IG = H(parent) - weighted H(children)

Pick the highest IG. Continuous features: test midpoints between sorted values.

Induction: a worked split

8 honey samples: 4 pure (P), 4 adulterated (A). Parent is a 4/4 mix, so **H(parent) = 1.00 bit.**

Split on a delta-13C isotope test:

branch	n	A	P	child H
yes	4	3	1	0.81
no	4	1	3	0.81

IG = 1.00 - 0.81 = **0.19 bit**

Why this split wins

A "dark colour?" split gives children that are each still 2/2, so H = 1.00 and IG = 0.00.

The isotope test carries more information, so it is chosen, then the tree recurses.

Stop by depth or leaf size; **prune** to fight overfitting.

Random forests

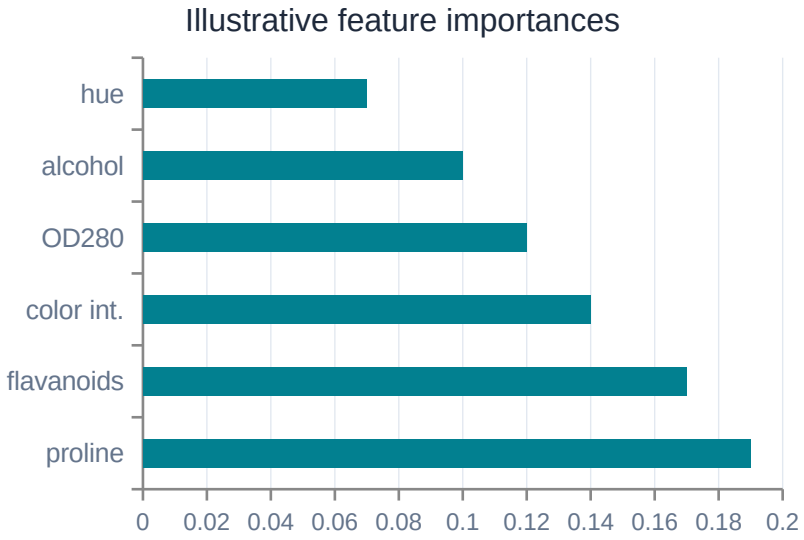
Many decision trees, averaged (or voted).

- a bootstrap sample of rows per tree
- a random subset of features per split

Averaging diverse, noisy trees cancels error.

+ accurate, robust, gives feature importances.

- less readable than one tree.



Example: flag likely genotoxic compounds from molecular descriptors; importance may highlight an aromatic nitro group.

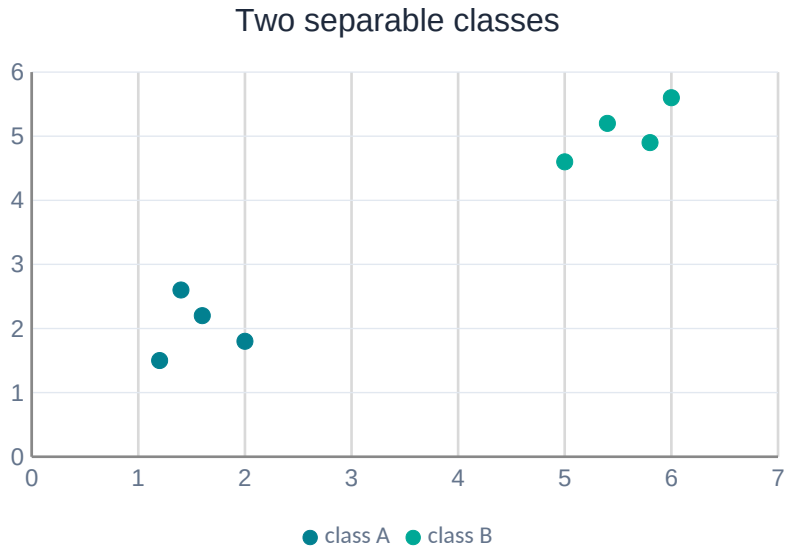
Support vector machines (and SVR)

Find the boundary with the widest margin between classes; the nearest points (support vectors) set it.

Kernel trick: map data to a higher space so a curved boundary becomes possible (RBF kernel).

SVR does the same for continuous targets.

Always scale features first.



Example: water above a contaminant threshold from conductivity (100s of uS/cm) and a trace metal (few ug/L): scale first.

k-means clustering (unsupervised)

No labels. Choose k, then repeat until stable:

- assign each sample to its nearest cluster centre
- move each centre to the mean of its members

Remember: choose k; results depend on the start; assumes round, similar-sized clusters; scale the features.



Example: cluster ICP-MS trace-element fingerprints of honey to probe floral or regional origin (scale first).

Choosing among them

Situation	Start with
Tabular data, strong result fast	Random forest
Interpretability is the priority	Single decision tree
Many features, few samples (scaled)	SVM / SVR
No labels, explore structure	k-means / clustering
Any situation	A trivial baseline, always

Deep learning, by intuition

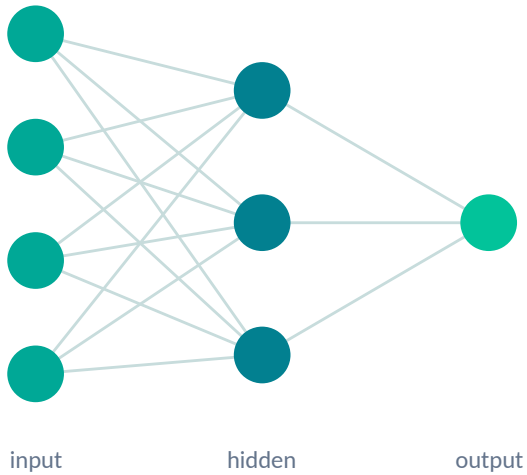
Neural networks - CNNs - transformers - (GNNs & autoencoders)

Neural networks: the foundation

Simple units (neurons): a weighted sum, an offset, then a non-linear activation.

Units form layers; output of one feeds the next. "Deep" = many layers. The non-linearity is essential.

Training: a loss measures error; backpropagation and gradient descent adjust weights over many epochs.



Example: learn a non-linear fluorescence calibration that stays accurate where a Beer-Lambert fit saturates.

Convolutional neural networks (CNNs)

The idea

A small filter slides across the input to detect a local motif wherever it occurs.

Early layers find simple motifs; later layers combine them. Few parameters, tolerant to shifts.

In analytical chemistry

Spectra are 1D sequences of intensities, so a 1D CNN learns peak shapes and patterns regardless of position.

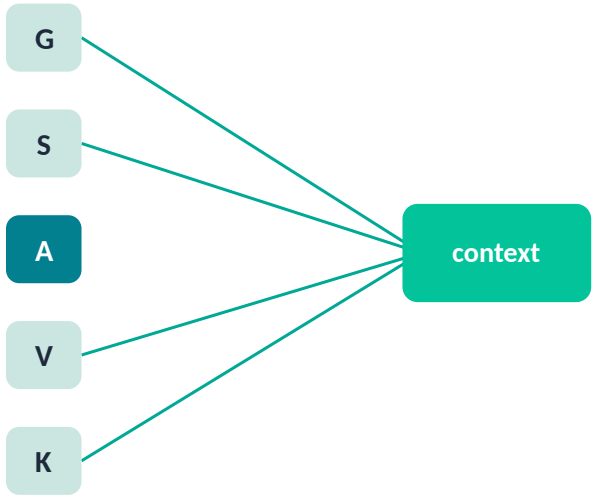
Example: classify a Raman spectrum as PE, PP or PS; the same idea picks true peaks against noise in LC-MS.

Transformers and attention

The architecture behind modern large language models, and increasingly models of molecules and spectra.

Attention: when processing one element, the model learns how much to attend to every other element, capturing long-range relationships.

Processed in parallel, scales to large data. Also the basis of the LLMs in Part 5.



one position attends to all others

Example: predict a peptide's fragmentation spectrum from its sequence (Prosit) using long-range residue relationships.

Two more to recognise (briefly)

Graph neural networks

Operate on graphs of nodes and edges. A molecule is a graph (atoms = nodes, bonds = edges), so GNNs predict properties from structure; each node exchanges information with its neighbours.

Autoencoders

Reconstruct the input through a narrow bottleneck, learning a compact representation. Used for dimensionality reduction, denoising and anomaly detection; a VAE can also generate new examples.

Examples: *GNN predicts CCS from a molecular graph; an autoencoder flags an anomalous process-NMR spectrum.*

WRAP-UP

Classical or deep? A practical rule

Reach for deep learning

- large, high-dimensional data
- spatial/sequential structure (images, spectra)
- enough labels and compute

Reach for classical

- tabular, modest-size data
- interpretability matters
- you need a strong result quickly

Next - TD1: train a decision tree, random forest, SVM and k-means on real wine-chemistry data.

KEY TAKEAWAYS

- ML fits functions to data; deep learning is ML with many-layered networks.
- Features and their quality usually matter more than the algorithm.
- A tree is induced by greedy splits chosen on information gain, then pruned.
- Start classical (random forest) and always beat a trivial baseline.
- Good scores on already-seen data prove almost nothing (Part 3).